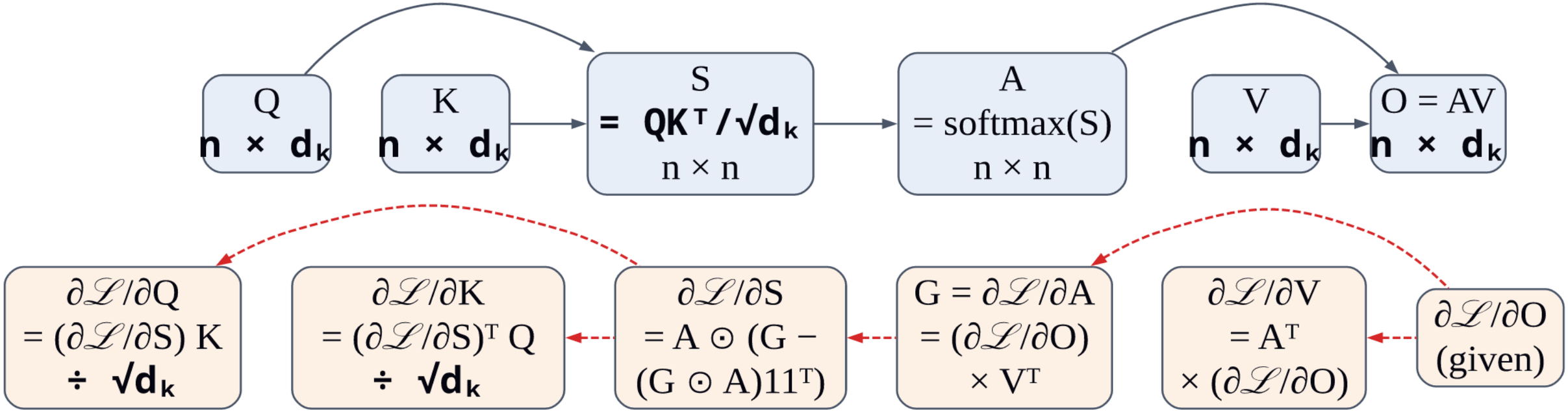

Q
n × d_k
K
n × d_k
S
= $QK^T/\sqrt{d_k}$
n × n
A
= softmax(S)
n × n
V
n × d_k
O = AV
n × d_k
$\partial\mathcal{L}/\partial Q$
= $(\partial\mathcal{L}/\partial S)$ K
÷ $\sqrt{d_k}$
$\partial\mathcal{L}/\partial K$
= $(\partial\mathcal{L}/\partial S)^T$ Q
÷ $\sqrt{d_k}$
$\partial\mathcal{L}/\partial S$
= A ⊙ (G −
(G ⊙ A)11^T)
G = $\partial\mathcal{L}/\partial A$
= $(\partial\mathcal{L}/\partial O)$
× V^T
$\partial\mathcal{L}/\partial V$
= A^T
× $(\partial\mathcal{L}/\partial O)$
$\partial\mathcal{L}/\partial O$
(given)